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Forecast Busts over Europe in Numerical Weather Prediction models: Characteristics and Variability

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Overview

This study is a part of the project that is investigating the mechanisms behind forecast busts. These are poor forecasts, that can occur due to unresolved and low-predictable processes in the numerical weather prediction model. Here, we compare bust responses for four leading operational centres: ECMWF, UKMO, NCEP and JMA.

Defining 'Forecast Busts'

A common way of defining forecast busts is in terms of errors in the prediction of the 500-hPa geopotential height (Z500), when the 6-day high resolution forecast of European Z500 has a root-mean-square error (RMSE) greater than 60 m and an anomaly correlation coefficient (ACC) less than 40% (Rodwell et al., 2013).

$$ACC = \frac{\sum_{i=1}^{N}\left[(Z500_f - Z500_c) - \overline{(Z500_f - Z500_c)}\right]\left[(Z500_a - Z500_c) - \overline{(Z500_a - Z500_c)}\right]}{\sqrt{\sum_{i=1}^{N}\left[(Z500_f - Z500_c) - \overline{(Z500_f - Z500_c)}\right]^2}\sqrt{\sum_{i=1}^{N}\left[(Z500_a - Z500_c) - \overline{(Z500_a - Z500_c)}\right]^2}};$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(Z500_f - Z500_a\right)^2}$$

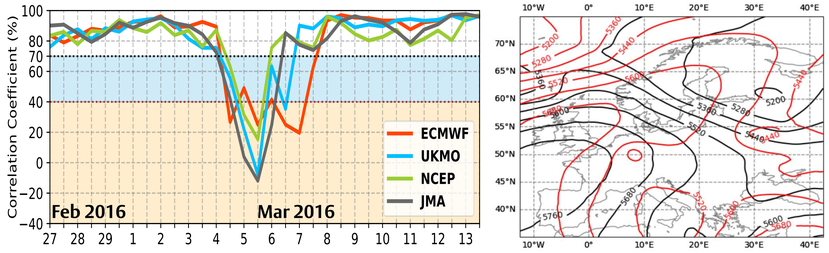


Fig. 1 (a) Time series of 6-day European Z500 ACC during Feb-Mar 2016 for 4 global NWP models. This includes the (b) bust case of 7th March 2016, 0000 UTC, where black lines are Z500 forecasts and red lines are Z500 analysis. The scores are calculated for 6-day forecasts over Europe (35°N–75°N, 12.5°W–42.5°E). ERA5 climatology from 2001-2020 is used to calculate the ACC.

Common Forecast Busts within NWP models

- The frequency of forecast busts in each model was analysed for the period 2009-2023, and the total counts are summarized Table 1.
- The forecast bust of 4th March 2016 (Fig. 1) demonstrates that models exhibit distinct behaviours before, during and after a bust event.
- These model responses motivated an analysis of how often busts coincide across models, with Table 1 also showing number of common busts between model pairs.
- Several busts occur independently across models, indicating variability in model performance and differences in how each centre represents the same atmospheric event.

	ECMWF	UKMO	NCEP	JMA
ECMWF	227			
UKMO	69	297		
NCEP	74	98	422	
JMA	66	78	101	378

Table. 1 Number of forecast busts occurring for each NWP model. The rows indicate number of busts for a centre given that the bust occurred in another centre (column wise).

Forecast Bust Periods

- We divided the ACC scale into three categories: good (ACC >70%), poor (ACC <=70%) and bust (ACC <40%).
- The whole bust period is represented by two distinct phases – Drop and Recovery – with the minimum ACC marking the transition point between them.
- The start and end of a bust episode are defined by the presence of three good forecasts preceding and following the episode.
- Based on this structure, several characteristics can be quantified, such as total duration, bust duration, drop rate and recovery rate (Fig 2).

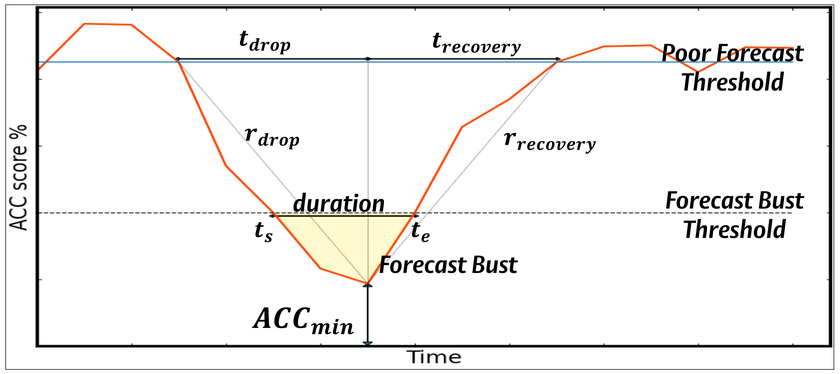


Fig. 2 Schematic of metrics used to define a forecast bust event. The figure shows the duration of a bust event which is defined by the time ACC remains below the threshold value, minimum ACC values during a bust event, start and end time of an event. Rate of drop and rate of recovery are calculated with reference to the poor forecast threshold (70%), time taken by the ACC to drop to the minimum and the time taken by the ACC to rise back to the poor forecast threshold.

Forecast Bust Profiles

A forecast bust profile represents the evolution of forecast skill during a bust, capturing the drop, minimum ACC and the recovery phases of model performance (Fig 3).

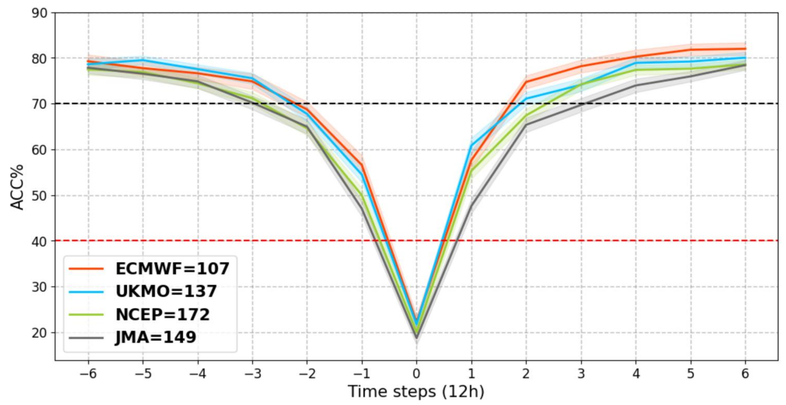


Fig. 3 Average ACC time series for all the identified forecast bust events for each of four NWP models over the period 2009-2023. Time 0 corresponds to the average of all the minimum ACC values achieved during a forecast bust period. The shading represents the standard error which is calculated at every time step by using bust period composite for each model.

Model-to-model comparison for bust characteristics

- ECMWF exhibits the least number of bust episodes and NCEP the highest.
- UKMO bust periods are the shortest and the ACC recovery is fastest.
- JMA has the longest duration of bust periods and the slowest drop and recovery rate. This points to the skill score remaining largely poor during these periods.
- NCEP exhibits slow drop during bust episodes but recovers significantly faster (Fig 3).

Future Work

- Forecast busts occur year-round, peaking in spring and autumn, indicating low predictability atmospheric events (Lillo and Parsons 2019).
- Mesoscale Convective Systems (MCSs) over US have linked to busts over Europe (Rodwell et al., 2013); ongoing work aims to assess how strongly high Convective Available Potential Energy (CAPE) correlates with busts.

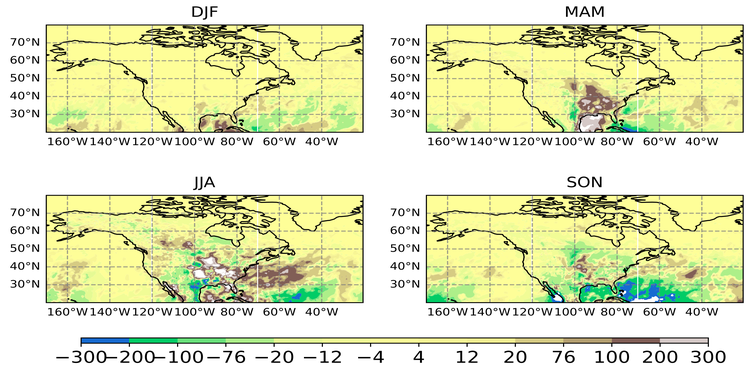


Fig. 4 Mean initial condition CAPE anomalies (J/kg) for ECMWF operational bust forecasts during 2009-2023 initialized at 0000 UTC for the four seasons (Winter, Spring, Summer and Autumn).

Conclusions

- Forecast bust profile analysis reveal that models exhibit similar structural behaviour during bust episodes, through differences in characteristics highlight model-specific responses to such events.
- Climatological comparisons of bust characteristics can help quantify how each model responds under similar conditions and across different seasons.
- Given that busts occur consistently, it is essential to understand the underlying mechanisms and how each model represents these processes differently to improve forecast reliability.

References

- Lillo, S.P. and Parsons, D.B., 2017. Investigating the dynamics of error growth in ECMWF medium-range forecast busts. Quarterly Journal of the Royal Meteorological Society.
- Rodwell, M.J., Magnusson, L., Bauer, P., Bechtold, P., Bonavita, M., Cardinali, C., Diamantakis, M., Earnshaw, P., Garcia-Mendez, A., Isaksen, L. and Källén, E., 2013. Characteristics of occasional poor medium-range weather forecasts for Europe. Bulletin of the American Meteorological Society.

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